

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B. Tech (CBCS) Degree Examinations
 Summer Semester July 2024

CC1104-1 – DATA COMMUNICATIONS
 (CC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- How many layers are there in the ISO OSI reference model?
 A) 7 B) 5
 C) 4 D) 6
- Which one of the following is not a function of network layer?
 A) Congestion control B) Data Compression
 C) Routing D) Inter-networking
- The layer responsible for encryption technique in data communication is ____
 A) Presentation B) Application
 C) Transport D) Session
- What was the name of the first network?
 A) ASAPNET B) ARPANET
 C) CNNET D) INTERNET
- In cyclic redundancy checking, what is the CRC?
 A) The divisor B) The Dividend
 C) The Remainder D) The Quotient
- Which error detection method consists of just one redundant bit per data unit?
 A) CRC B) CheckSum
 C) Hamming Distance D) Simple Parity
- QPSK system uses a phase shift of ____
 A) π B) $\pi/2$
 C) $\pi/4$ D) 2π
- In Alternate Mark Inversion (AMI) ____
 A) 0 is encoded as positive pulse and 1 is encoded as negative pulse B) 0 is encoded as no pulse and 1 is encoded as negative pulse
 C) 0 is encoded as negative pulse and 1 is encoded as positive pulse D) 0 is encoded as no pulse and 1 is encoded as positive or negative pulse
- Which multiplexing technique used to transmit digital signals?
 A) FDM B) TDM
 C) WDM D) FDM & WDM
- In ____, if device has no data to send, its time slot remains empty
 A) Synchronous TDM B) Asynchronous TDM
 C) Statistical TDM D) None of these
- In Go-Back-N ARQ, if 5 is the number of bits for the sequence number, then the maximum size of the receive window must be ____.
 A) 1 B) 15
 C) 16 D) 31
- In.....ARQ, if a NAK is received, only the specified damaged or lost frame is transmitted.
 A) Go-Back-N B) Selective Repeat
 C) Stop-and-wait D) All of these

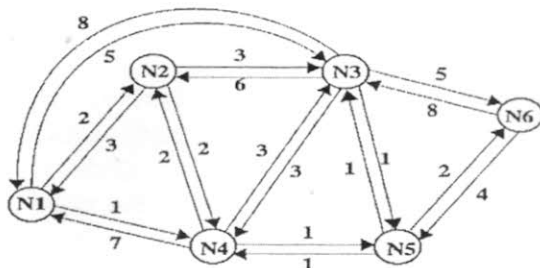
13. The required resources for communication between end systems are reserved for the duration of the session between end systems in _____ method.
- A) Packet switching B) Circuit switching
C) Line switching D) Message Switching
14. The time slot interchange uses _____ multiplexing to achieve switching
- A) FDM B) Switch division
C) TDM D) WDM
15. A _____ switch usually has a higher efficiency than a _____ switch
- A) Single stage; multistage B) Single crossbar; multistage
C) Crossbar; dual D) Multistage; single crossbar
16. Which layer is responsible for determining routes?
- A) Physical Layer B) Transport Layer
C) Network Layer D) Transmission Layer
17. Thebucket algorithm allows idle hosts to accumulate credit for the future in the form of tokens.
- A) Leaky B) Token
C) Both A & B D) None of these
18. Which multiple access technique is used by IEEE 802.11 standard for wireless LAN?
- A) CDMA B) CSMA/CA
C) ALOHA D) CSMA/CD
19. In wireless network an extended service set is a set of _____
- A) connected basic service sets B) all stations
C) all access points D) connected access points
20. What is start frame delimiter (SFD) in ethernet frame?
- A) 10101010 B) 10101011
C) 00000000 D) 11111111

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the various Internet terminologies, with neat Internet architecture diagram.	6	L*2	1	1
	b) Explain the digital signal encoding method adopted in NRZ-L and NRZ-I encoding technique with suitable example. Mention their advantages and disadvantages.	6	L2	2	1
	c) Define attenuation. Which device is used to strengthen the attenuated signal? Suppose a signal travels through a transmission medium and its power is reduced to one-half. Measure the loss in decibel.	4	L1	2	1,2
2.	a) Define scrambling. Explain B8ZS based on a bipolar-AMI with suitable example.	6	L2	2	1,2
	b) Suppose that the spectrum of a channel is between 3MHz and 4MHz and $SNR_{dB} = 24dB$. Find the following: i.) SNR, ii.) Capacity of the channel iii.) How many signal levels are required?	6	L1	2	1,2
	c) Define Hamming Distance. Find the Minimum Hamming Distance for following valid codewords 011011, 10101, 11110.	4	L1	2	1,2
3.	a) Define Protocol. Explain protocol architecture in networks with suitable diagram.	6	L2	1	1,2
	b) Summarize the differences between ASK and FSK.	4	L2	2	1,2
	c) Apply traditional internet checksum to find the checksum for the following set of data (0001) ₁₆ (F203) ₁₆ (F4F5) ₁₆ (F6F7) ₁₆ . Show how receiver verifies the checksum.	6	L3	2	1,2

Unit – II

4. a) What is the need of flow control? Explain Stop-and-Wait ARQ protocol for flow and error control. 6 L2 3 1,2
 b) Define Spread Spectrum. Explain DSSS. 4 L2 3 1,2
 c) Describe Time Division Switching. How Time Slot Interchange mechanism works? Explain. 6 L2 4 1,2
5. a) Consider four sources apply TDM, each creating 250 characters per second. If the interleaved unit is a character and 1 synchronizing bit is added to each frame, find (i) the data rate of each source, (ii) the duration of each character in each source, (iii) the frame rate, (iv) the duration of each frame, (v) the number of bits in each frame, and (vi) the data rate of the link. 6 L1 3 1,2
 b) Compare Packet switching and Circuit switching. 4 L2 4 1,2
 c) Apply Bellman-Ford algorithm to find the shortest path from source node 'N1' in the following network graph. Draw the final spanning tree



6. a) Explain Wavelength Division Multiplexing technique. 4 L2 3 1,2
 b) With frame format of HDLC protocol, explain each fields of the frame. 6 L2 3 1,2
 c) List advantages of packet switching over circuit switching. Explain Virtual Circuit Switching technique with suitable diagram. 6 L2 4 1,2
- Unit – III**
7. a) Briefly explain specifications of Fast Ethernet. Summarizes key characteristics of the 100BASE-T options. 6 L2 5 1,2
 b) What is the need of congestion control? Explain Implicit Congestion Signaling mechanism in congestion control. 4 L2 5 1,2
 c) With a flowchart, explain IEEE 802.11 Medium Access Control Logic 6 L2 5 1,2
8. a) Explain the CSMA/CD operation. 8 L2 5 1,2
 b) With a neat frame format explain each fields of IEEE 802.3 MAC Frame. 8 L2 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
